

Glomerular (Nephrotic) — Membranous Nephropathy



1. White adults #1

WHAT YOU SEE

Sign 'WHITE ADULTS #1' on right tower

WHAT IT ENCODES

Membranous nephropathy (MN) is the most common cause of PRIMARY nephrotic syndrome in white/Caucasian adults, peaking in the 4th–6th decades with a male predominance. (For comparison: FSGS is the most common primary cause overall in US adults and especially in Black/Hispanic patients, while minimal change disease dominates in children.) Presents with insidious-onset full nephrotic syndrome: proteinuria >3.5 g/day, hypoalbuminemia, edema, and hyperlipidemia.

BOARD TRAP

WRONG: 'Minimal change disease' is the most common nephrotic cause in this adult patient. Discriminator — MCD is the #1 nephrotic cause in CHILDREN (age 2–6) and is steroid-responsive within days; in a white adult with insidious nephrotic-range proteinuria the answer is membranous, confirmed by anti-PLA2R and subepithelial spikes.

2. Anti-PLA2R 70-80%

WHAT YOU SEE

Stained-glass saint 'PLA2R 70-80%'

WHAT IT ENCODES

Anti-phospholipase A2 receptor (anti-PLA2R) IgG4 autoantibodies are present in ~70–80% of PRIMARY (idiopathic) membranous nephropathy and are the defining serologic marker. PLA2R is an antigen on the podocyte surface; circulating antibody binds in situ to form subepithelial immune complexes. Titer tracks disease activity — falling titer predicts remission, rising/persistent titer predicts relapse or treatment failure. A smaller subset is anti-THSD7A positive (~3%, associated with malignancy).

BOARD TRAP

WRONG: 'Anti-GBM antibody' is the marker for membranous. Discriminator — anti-GBM defines Goodpasture/anti-GBM disease (linear IF, crescentic RPGN, lung hemorrhage), a NEPHRITIC process. Membranous is marked by anti-PLA2R with GRANULAR subepithelial IgG, not linear staining.

3. Primary = subepithelial only

WHAT YOU SEE

Banner 'PRIMARY = SUBEPITHELIAL ONLY / IgG + C3'

WHAT IT ENCODES

Primary MN shows immune deposits located ONLY in the subepithelial space (between podocyte and GBM), with granular capillary-loop staining for IgG (predominantly the IgG4 subclass) and C3 on immunofluorescence. Because deposits sit outside the GBM away from circulating cells, there is little inflammation/proliferation — hence a 'bland' nephrotic (not nephritic) picture. PLA2R antigen staining on biopsy further supports a primary etiology.

BOARD TRAP

WRONG: 'Subendothelial deposits' define primary membranous. Discriminator — subENDOthelial deposits (inner side of GBM) point to MPGN or lupus class III/IV (tram-tracks, low complement, nephritic features). Primary MN deposits are strictly subEPithelial. Subendothelial/mesangial deposits in an MN biopsy suggest a SECONDARY cause (e.g., lupus class V).

4. Spike-and-dome

WHAT YOU SEE

'SPIKE + DOME' label at top of window

WHAT IT ENCODES

Jones methenamine SILVER stain shows the classic 'spike-and-dome' pattern: argyrophilic GBM spikes (new basement-membrane material) projecting up between the non-staining subepithelial immune 'domes.' Over time the spikes extend and surround the deposits, thickening the GBM diffusely without hypercellularity. This is the light-microscopy correlate of subepithelial deposition.

BOARD TRAP

WRONG: 'Tram-track double contours' are seen in membranous. Discriminator — tram-tracking (GBM splitting/double contours from mesangial interposition) is the hallmark of MPGN, not MN. Spike-and-dome on silver stain = membranous; tram-tracks = MPGN.

5. Foot process effacement

WHAT YOU SEE

'FOOT PROCESS EFFACEMENT' on bench

WHAT IT ENCODES

Electron microscopy shows electron-dense SUBEPITHELIAL deposits with diffuse effacement (fusion/flattening) of the overlying podocyte foot processes — the structural basis of the heavy proteinuria. Diffuse foot-process effacement is common to ALL nephrotic diseases; what localizes the diagnosis to MN is the subepithelial position of the deposits.

BOARD TRAP

WRONG: EM shows 'mesangial deposits with mesangial proliferation' in membranous. Discriminator — mesangial deposits with proliferation describe IgA nephropathy or lupus; MN deposits are subepithelial. Note effacement alone is non-specific (also in MCD/FSGS) — the deposit LOCATION is the key.

6. Anti-PLA2R → no biopsy

WHAT YOU SEE

Plaque on the bench reading 'ANTI-PLA2R = NO BIOPSY' — the message: a positive anti-PLA2R with a typical nephrotic picture and normal eGFR lets you diagnose primary MN without biopsying.

WHAT IT ENCODES

In a patient with nephrotic syndrome, preserved/near-normal kidney function, and a positive anti-PLA2R antibody, the diagnosis of primary MN can be made WITHOUT a renal biopsy (per KDIGO). Biopsy is still pursued when serology is negative, function is declining, the picture is atypical, or a secondary cause must be excluded.

BOARD TRAP

WRONG: 'A renal biopsy is mandatory even when anti-PLA2R is positive.' Discriminator — modern KDIGO guidance allows a serology-based diagnosis when anti-PLA2R is positive with a compatible clinical picture and normal eGFR; reflexively biopsying every PLA2R-positive patient is outdated. Biopsy is reserved for negative serology or atypical/declining cases.

7. Foot process effacement (EM dense deposits)

WHAT YOU SEE

EM panel near 'FOOT PROCESS EFFACEMENT' bench detail

WHAT IT ENCODES

NEW high-yield add: Ehrenreich–Churg staging on EM grades MN progression — I: small subepithelial deposits; II: spikes between deposits; III: deposits incorporated into thickened GBM; IV: deposits resorbed leaving lucent GBM. Later stages correlate with chronicity/scarring rather than active disease.

BOARD TRAP

WRONG: 'Higher Ehrenreich–Churg stage always means worse active disease needing more immunosuppression.' Discriminator — advanced stages reflect chronicity (deposits being resorbed), not necessarily active immunologic activity; treatment is guided by clinical risk and PLA2R titer, not stage alone.

8. Secondary causes

WHAT YOU SEE

Left wall 'SECONDARY + EXTRA SITES + IgG1/2/3' — syphilis, donor, breast/lung tumor

WHAT IT ENCODES

Secondary MN mimics primary but is driven by an external trigger. Key causes — solid-organ MALIGNANCY (lung, breast, colon, prostate; suspect in patients >65), infections (hepatitis B, syphilis), autoimmune (SLE class V), and DRUGS (NSAIDs, gold, penicillamine, captopril). Clues to secondary disease: deposits at 'extra' sites (mesangial/subendothelial), IgG1/IgG2/IgG3 rather than IgG4, full-house IF, and negative anti-PLA2R. Always age-appropriate cancer screening in older adults.

BOARD TRAP

WRONG: 'All membranous nephropathy is primary/idiopathic, so no further workup is needed.' Discriminator — ~25% of MN is secondary. Missing an underlying solid tumor, hepatitis B, lupus, or a culprit drug (NSAIDs, gold) is the classic error; secondary clues are IgG1/2/3 subclass, mesangial/subendothelial deposits, and negative PLA2R.

9. Lupus → full house

WHAT YOU SEE

Red 'unless lupus' tag near C3/C4 dial

WHAT IT ENCODES

Lupus membranous (ISN/RPS class V) is a SECONDARY MN. Unlike primary MN (normal complement), lupus class V shows low C3 AND low C4 (classical-pathway consumption) and 'full-house' immunofluorescence — IgG, IgA, IgM, C3, AND C1q all positive. Tubuloreticular inclusions on EM ('interferon footprints') further point to lupus.

BOARD TRAP

WRONG: 'Complement is always normal in membranous, so low C3/C4 rules it out.' Discriminator — complement is normal in PRIMARY MN, but lupus class V (secondary MN) consumes complement (low C3/C4) and shows full-house IF with C1q. Low complement + full house in a membranous pattern = lupus until proven otherwise.

10. RVT #1 / thrombosis risk

WHAT YOU SEE

Gargoyle 'RVT #1', DVT, PE, 'ALB ↓ = ↑ RISK'

WHAT IT ENCODES

MN carries the HIGHEST venous thromboembolism risk of all nephrotic syndromes — classic complication is RENAL VEIN THROMBOSIS (flank pain, hematuria, acute LDH rise, asymmetric kidneys/left-sided varicocele), plus DVT and PE. Mechanism: urinary loss of antithrombin III and other anticoagulant proteins plus a relatively preserved synthesis of procoagulants → net hypercoagulable state. Risk rises sharply when serum albumin falls below ~2.5–2.8 g/dL; prophylactic anticoagulation (e.g., warfarin) is considered for high VTE risk.

BOARD TRAP

WRONG: 'Membranous carries no clotting risk; new flank pain and hematuria must be a kidney stone.' Discriminator — in a nephrotic MN patient, acute flank pain + hematuria + LDH rise is RENAL VEIN THROMBOSIS until proven otherwise. MN has the highest thrombotic risk of the nephrotic syndromes, driven by antithrombin III loss; image/anticoagulate, don't dismiss.

11. Rituximab (anti-CD20)

WHAT YOU SEE

Friar 'RITUXIMAB' / 'RITUX → IgG4'

WHAT IT ENCODES

Rituximab (anti-CD20 monoclonal antibody) depletes B cells, cutting off production of the pathogenic anti-PLA2R (IgG4) antibody, and is now FIRST-LINE immunosuppression for high-risk primary MN (e.g., MENTOR/GEMRITUX data). Typical dosing is 1 g IV on days 0 and 14, or 375 mg/m² weekly ×4. Remission tracks with disappearance of anti-PLA2R antibody.

BOARD TRAP

WRONG: 'Steroids alone cure most membranous nephropathy.' Discriminator — corticosteroid MONOTHERAPY is ineffective in MN (unlike minimal change disease, which is exquisitely steroid-responsive). Effective regimens are rituximab or steroid + cyclophosphamide (Ponticelli) or a calcineurin inhibitor; do not pick steroid monotherapy.

12. Cyclophosphamide + steroids

WHAT YOU SEE

Scroll labelled 'PONTICELLI — STEROIDS + CYCLOPHOSPHAMIDE' on the cathedral wall — the paired regimen named on the scroll is the alternating monthly steroid/cyclophosphamide protocol for high-risk MN.

WHAT IT ENCODES

The modified PONTICELLI regimen alternates monthly cycles of corticosteroids (IV methylprednisolone pulses then oral prednisone) with oral cyclophosphamide over 6 months — a proven option for high-risk primary MN that lowers ESKD risk. Alternatives include calcineurin inhibitors (cyclosporine/tacrolimus). Cyclophosphamide toxicities to counsel on: hemorrhagic cystitis (give mesna/hydration), infertility, infection, and bladder cancer risk.

BOARD TRAP

WRONG: 'Pure steroid monotherapy is the standard of care to induce remission in MN.' Discriminator — MN does not respond to steroids alone; the validated immunosuppressive regimens are rituximab or the Ponticelli steroid+cyclophosphamide protocol (or a CNI). Steroid monotherapy is the trap answer carried over from minimal change disease.

13. Risk-based therapy

WHAT YOU SEE

Traffic-light tower: LOW=watch, MOD=reassess, HIGH=start IS, VERY HIGH=urgent

WHAT IT ENCODES

Treatment is RISK-stratified (KDIGO), not automatic. LOW risk (normal eGFR, proteinuria <3.5 g, albumin >3): conservative — RAAS blockade (ACEi/ARB), salt restriction, statin, diuretics, and observe 6 months for spontaneous remission. MODERATE: reassess at 6 months. HIGH/VERY-HIGH risk (declining eGFR, proteinuria >8 g, low albumin, high/rising anti-PLA2R, life-threatening complications): start immunosuppression (rituximab or Ponticelli) promptly.

BOARD TRAP

WRONG: 'Start immunosuppression in every MN patient immediately at diagnosis.' Discriminator — up to one-third of MN remits spontaneously, so low-risk patients are managed conservatively with RAAS blockade and observation. Reflexive immunosuppression in a low-risk patient exposes them to needless toxicity; treatment follows the KDIGO risk tiers.

14. 1/3 remit, 1/3 relapse, 1/3 ESKD

WHAT YOU SEE

Bottom captions '1/3 REMIT', '1/3 RELAPSE', '1/3 ESKD'

WHAT IT ENCODES

Classic natural-history 'rule of thirds' for untreated primary MN: roughly 1/3 undergo spontaneous remission, 1/3 have persistent proteinuria with relapsing/remitting course, and 1/3 progress to ESKD over 5–10 years. This is precisely why low-risk patients are observed before committing to immunosuppression.

BOARD TRAP

WRONG: 'Nearly all cases of membranous resolve spontaneously without treatment.' Discriminator — only ~1/3 spontaneously remit; the other two-thirds relapse or progress, and high-risk patients march toward ESKD without immunosuppression. Overstating spontaneous remission leads to under-treating high-risk disease.

15. Interstitial fibrosis = prognosis

WHAT YOU SEE

'INTERSTITIAL FIBROSIS = PROGNOSIS'

WHAT IT ENCODES

As in essentially all chronic glomerular disease, the degree of TUBULOINTERSTITIAL FIBROSIS and tubular atrophy on biopsy — not the glomerular deposit severity — is the strongest predictor of long-term renal outcome. Other adverse prognostic markers in MN: male sex, older age, higher/persistent proteinuria, reduced eGFR, and high anti-PLA2R titer.

BOARD TRAP

WRONG: 'Glomerular deposit severity (spike-and-dome extent) is the best predictor of prognosis.' Discriminator — across glomerulonephritides the best prognostic marker is the extent of tubulointerstitial fibrosis/tubular atrophy, because fibrosis is irreversible. Grading glomerular deposits does not predict ESKD as well as the interstitial scarring does.